MIL-S-51244A(EA)
26 February 1976
SUPERSEDING
MIL-S-51244(MU)
29 June 1967

MILITARY SPECIFICATION

SERVICE KIT, PORTABLE FLAME THROWER,

RIOT CONTROL AGENT DISPERSER, M27

This specification is approved for use by all
Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of service kit (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-C-843 - Cushioning, Material, Cellulosic.
PPP-P-291 - Paperboard, Wrapping, Cushioning.
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.

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MIL-P-130 - Paper, Wrapping, Laminated and Creped,
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification For.

STANDARDS

FEDERAL

Fed. Std. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods.

DRAWINGS

US ARMY ARMAMENT COMMAND

D81-6-369 - Service Kit, Portable Flame Thrower, Riot Control Agent Disperser, M27.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D81-6-369 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings-

3.1.2 Components. All components of the kit shall conform to the specifications and drawings listed on Drawing D81-6-369 and subsidiary drawings.

3.2 Assembly . The service kit shall be assembled in accordance with Drawing D81-5-36.

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3.3 Regulator, fluid pressure. The fluid pressure regulator shall not leak and shall be capable of regulating air from 0 to 40 pounds per square inch gage (psig) maximum when subjected to an air pressure of 2000 ± 50 psig for a minimum period of 6 minutes when tested as specified in 4.3.

3.4 Workmanship. The service kit shall be manufactured and assembled in accordance with standard commercial practice. The kit shall be free from contamination (foreign matter, oil or grease) and damage such as dents or deformation that could affect its intended use (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of like components, produced by one manufacturer, from the same materials, under essentially the same manufacturing conditions.

4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Examination. Sample components and level A packing shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.4 Critical defects. Each component shall be inspected for the critical characteristics listed in the classification of defects.

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4.2.5 Classification of defects.

(a) Service kit, portable flamethrower, RCA disperser, M27 (Dwg D81-6-369)

Categories	Defects
Critical:	None defined
Major:	100% inspection
101	Components missing
102	List of contents missing
103	Marking missing, incorrect or illegible
104	Workmanship (3.4)

(b) Chest, packing (Dwg D81-6-78)

Categories	Defects
Critical.:	None defined.
Major:	AQL 1.0 percent defective
101	Components missing, loose or incorrectly assembled
102	Inside length, width or height incorrect
103	Chest tray length, height or width incorrect (Dwg D81-6-81)
104	Partitions incorrectly located (Dwg D81-6-81)
105	Workmanship (3.4)
Minor:	AQL 2.5 percent defective
201	Protective finish missing

(c) Regulator, fluid Pressure (Dwg L/M81-6-369)

Categories	Defects	Acceptance standards
Critical:		
1	Leakage and functioning	4.3

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Major: AQL 1.0 percent defective

101	Outlet connection thread incorrect
102	Inlet connection thread incorrect
103	Gage range incorrect
104	Workmanship (3.4)

(d) Preparation for delivery (section 5)

Categories	Defects
Critical:	None defined

Major: AQL 1.0 percent defective

101	Components missing
102	Components not cleaned and dried as specified
103	Preservatives missing or incorrect
104	Packaging methods incorrect
105	Protective caps, plugs or sleeves missing
106	Container, box or package damaged
107	Marking missing, incorrect or illegible
108	Strapping loose or missing

4.3 Pressure regulator leakage and functioning test.

a. Connect outlet port to a pressure gage and a shut-off valve.

b. Connect the inlet port to a controlled source of air pressure.

c. Close the shut-off valve and subject the regulator to 2000 ± 50 psig. Record the outlet pressure.

d. Apply a suitable leak detector solution to all joints and observe for leakage.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.3).

5.1 Cleaning and drying. All components of the kit including the tools and chest shall be cleaned and dried in accordance with the applicable provisions of MIL-P-116. Cleaning method C-1 and any of the drying methods listed shall be used,

5.2 Preservation and packaging, levels shall be as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Preservatives and packaging methods. Unless otherwise specified, the preservatives including the application thereof and packaging methods cited herein shall be in conformance with the requirements of MIL-P-116.

5.2.1.2 Processing of components except tools and publications. Unpainted ferrous and aluminum surfaces of components used in compressed air systems below 200 psi and the fuel systems shall be coated with type P-7 or P-10, grade 2 or 3 preservative oil. Unpainted surfaces of components used in compressed air systems in excess of 200 psi shall not be coated with a preservative. Unpainted nonferrous surfaces, except aluminum surfaces, shall not be coated with a preservative. Plastic caps, plugs or sleeves, fabricated in accordance with MIL-C-5501, shall be used to protect male threads on all components of the kit and to close off all openings and ports of the pressure regulator and valves.

5.2.1.3 Pressure regulator with gag (FSN 6680-494-5229). The pressure regulator with gage shall be preserved and pack accordance with method IA-8 and placed within a weather-resistant fiberboard (W6c) box conforming to PPP-B-636. Cushioning material conforming to type II, class B of PPP-C-843 shall be as required to prevent the item from moving within the container. The following listed components of the pressure regulator shall be packaged together in accordance with method 111. Each item shall be wrapped using barrier material conforming to type II, grade A, class 2 of MIL-B-121 and secured by PPP-T-76 tape. The wrapped items shall be placed in a box conforming to style RSC, grade W6c of PPP-B-636.

(a) Adapter, straight, pipe-to-tube (Dwg B81-6-141).
(b) Adapter, straight, pipe-to-tube, (Dwg B81-6-405).
(c) Adapter, (Dwg C81-6-150)
(d) Coupling half, quick disconnect (Dwg B81-1-884).
(e) Screw, valve adjusting, brass (removed component of pressure regulator) optional, depending on type of regulator supplied.

5.2.1.4 Filling-line assembly. The filling line (Dwg D81-3-4) shall be individually wrapped using barrier material conforming to type II, grade A, class 2 of MIL-B-121. The wrapped items shall be packaged together using method IA-14. Boxes shall be style RSC, grade W6c of PPP-B-636.

5.2.1.5 Valve; gate, brass and 3/4 by 4 inch brass pipe-nipple. The gate and nipple shall be separately wrapped in barrier material. conforming to type II, grade A, class 2 of MIL-B-121 and packaged together using method IC-1.

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5.2.1.6 Test gage assemblies. The four test gages (Dwg C81-6-253, C81-6-379, B81-6-377, and C81-6-90) shall be individually packaged in accordance with method IA-15. The boxes shall conform to class weather-resistant, grade W6C of ppp-b-636.

5.2.1.7 Hose assemblies, 6, 8 and 15 foot. The hose fittings shall be cushion wrapped with material conforming to type II, grade A, class 2 of MIL-B-121 or type II of MIL-P-130. Each hose assembly shall be coiled with minimum of bending that will permit the placement of the coiled hose within the chest provided. Before placement in chest, each coil shall be secured by tying or taping.

5.2.1.8 Adapter, brass, pressure cylinder, types C and D (Dwg D81-6-263). Each adapter shall be packaged method III, by wrapping in cushioning material conforming to type II, class B, of PPP-C-843 and securing with tape. The wrapped items shall be placed in a style RSC, grade W6C fiberboard container conforming to PPP-B-636.

5.2.1.9 Adapter, steel, with quick release coupling. The adapter (Dwg B81-6-385) shall be packaged in accordance with method IC-1).

5.2.1.10 Bushing, pipe, steel, 2 inch and 3/4 inch. The bushing shall be wrapped in cushioning material conforming to type II, class B of PPP-C-843, packaged method IC-1 and overpacked in a style RSC, grade W6c container conforming to PPP-B-636.

5.2.1.11 Key, socket head, screw, 5/32 inch. The blade shall be packaged in accordance with method IC-1 or IC-3.

5.2.1.12 Pliers, slip-joint and adjustable hook spanner (Dwg C81-1-9726). The joints on each item shall be coated with P-7 type preservative and operated to insure penetration. Items shall then be wrapped in type II, grade A, class 2 barrier material conforming to MIL-B-121 and packaged method IC-1 or IC-3.

5.2.1.13 Screwdriver and offset screwdriver (Dwg D81-1-9727). The screwdrivers shall be packaged method IC-1 or IC-3.

5.2.1.14 Wrenches, adjustable. The 8 inch, 9 inch and 15 inch adjustable wrenches shall have type P-7 preservative oil applied to the adjusting screw. The screw shall be operated to assure penetration of the oil. The wrenches shall be wrapped in type II, grade A, class 2, barrier material conforming to MIL-B-121. The wrenches shall be individually packaged method IC-1 or IC-3.

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5.2.1.15 Wire, steel, 1/4 pound coil. The coil of wire shall be coated with type P-7 preservative oil and packaged in accordance with method IC-1.

5.2.1.16 Technical publications. The publications shall be packaged in accordance with either method IC-1 or IC-3.

5.2.1.17 Reducers, tube (Dwg B81-3-61,- B81-6-306) body check valve (Dwg B81-1-2147) and test adapter (Dwg C81-1-9719). These items shall be individually wrapped in type II, grade A, class 2 barrier material conforming to MIL-B-121, and packaged method IC-1 or IC-3.

5.2.1.18 Chest (Dwg D81-6-78), The chest hinges and hasps shall be oiled with type P-7 preservative oil and these components shall be actuated to assure penetration of the oil. All excess oil shall be wiped off.

5.2.2 Commercial packaging. The components of the kit, including the technical publications, shall be packaged in accordance iwth Fed. Std. 356.

5.3 Packing, level A. The components of the kit, tools and technical publications preserved and packaged as specified in 5.2.1.1 through 5.2.1.17 shall be packed within the chest specified in 5.2.1.18 (DWg D81-6-78). Internal cushioning, blocking or bracing hsall be applied as required in accordance with MIL-STD-1186, to prevent movement of the packaged items within the chest. The chest shall be closed and strapped in accordance with the appendix of PPP-B-601. Galvanized strapping shall be used.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MHAYl!D-129 to include lot number and shipping containers shall be marked in accordance with Drawing D81-6-419.

6. NOTES

6.1 Intended use. The kit covered by this specification is intended for servicing in the field the M2A1-7 portable flame thrower, ABC M9-7, ABC M9A1-7, portable flame thrower, M3 irritant gas disperser and M4 irritant gas disperser. The kit is used to charge air cylinders and fill fuel tanks on the portable flame thrower and dispersers. The M27 service kit consists of the components (Dwg D81-6-369) packed in a wood chest (component as used in this specification shall be understood to mean part, subassembly or assembly as applicable). The service kit is used with a pressure source to charge pressure cylinders, to test and adjust pressure regulators of portable flame throwers and riot control agent dispersers and to charge commercial air cylinders. The service kit is also used to transfer thickened fuel or unthickened fuel from 55-gallon drums to portable flame throwers, fire bombs or flame land mines.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging required.

6.3 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the kit should be used. (Copies of repair parts packaging data sheets may be obtained from the Commander, Edgewood Arsenal, Attn: SAREA-TS-D, Aberdeen Proving Ground, MD 21010.)

6.4 Supplementary quality assurance provisions (SQAP).

SQAPS for the M27 Service Kit are:

SQAP 81-1-2147, Body, Check Valve
SQAP 81-3-4, Line Assembly, Filling
SQAP 81-3-6, Hose Assembly, Metal
SQAP 81-3-30, Adapter, Straight, Pipe-To-Tube
SQAP 81-3-61, Reducer, Tube
SQAP 81-3-66, Hose Assembly
SQAP 81-6-90, Test Gage Assembly. Pressure Cylinder
SQAP 81-6-141, Adapter Straight, Pipe-To-Tube
SQAP 81-6-150, Adapter
SQAP 81-6-151, Hose Assembly, Rubber
SQAP 81-6-253, Test Gage Assembly, Fuel Tank (M2Al)
SQAP 81-6-254, Adapter, Testing Gage
SQAP 81-6-263, Adapter, Pressure Cylinder
SQAP 81-6-306, Reducer, Tube
SQAP 81-6-377, Test Gage Assembly, Agent Tank (M3)
SQAP 81-6-379, Test Gage Assembly, Fuel Tank (M9)
SQAP 81-6-380, Adapter, Testing Gage
SQAP 81-6-384, Adapter, Straight, Pipe-To-Tube
SQAP 81-6-405, Adapter, Straight, Pipe-To-Tube
SQAP 81-6-408, Hose Assembly

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A129

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH -POUND

MIL-S-51244A(EA)
NOTICE 1
28 September 1995

MILITARY SPECIFICATION

SERVICE KIT, PORTABLE FLAME THROWER RIOT CONTROL AGENT DISPERSER, M27

This notice should be filed in front of MIL–S-51244A(EA) dated 26 February 1976.

MIL–S–51244A(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army – EA

Project No. 104O-A216

AMSC N/A

FSC 1040

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